
Data Structures

BS (CS) _Fall_2024

Lab_14 Tasks



Learning Objectives:

1. Hashing

Lab Tasks

Submission Instructions

1. Create a separate **.h** and **.cpp** file for each class.
2. Now create a new folder with name *ROLLNO_SEC_LAB14* e.g. *i23XXXX_A_LAB07*(Also name your cpp file like this.)
3. Move all of your files to this newly created directory and compress it into a **.zip file**. Return the test file as well.
4. Now you have to submit this zipped file on Google Classroom.
5. If you don't follow the above-mentioned submission instruction, you will be marked **zero**.
6. Plagiarism or AI generated code in the Lab Task will result in **zero** marks in the whole category.

Question 1

Write a C++ code to implement 'Chaining' in a Hash Tabel. You will need to make the following classes:

- Class Node {
 int key
 Int value
 Node *next;
 }
 You will need to make the constructor of this class as well.
- Class HashTable {
 Int capacity
 Node ** arr_of_linkedlist
 }

• Int Hash_Function(int key) // You can use the division hash function
• Void insert (int key , int value)
• Void remove (int key)
• Int search()
• ~HashTable()

Question 2

In this task, you are asked to make insertions in the table using **Quadratic probing**. Suppose that an item with key X is hashed at t, that is, $h(X) \% t$ and $0 \leq t < \text{HTSize} - 1$.

Further suppose that position t is already occupied. In quadratic probing, starting at position t, we linearly search the array at locations $(t + 1) \% \text{HTSize}$, $(t + 2) \% \text{HTSize}$ $(t + 3) \% \text{HTSize}$, $(t + 4) \% \text{HTSize}$, $(t + 5) \% \text{HTSize}$, $(t + 6) \% \text{HTSize}$, $(t + 7) \% \text{HTSize}$, $(t + 8) \% \text{HTSize}$, $(t + 9) \% \text{HTSize}$, $(t + 10) \% \text{HTSize}$, $(t + 11) \% \text{HTSize}$, $(t + 12) \% \text{HTSize}$, $(t + 13) \% \text{HTSize}$, $(t + 14) \% \text{HTSize}$, $(t + 15) \% \text{HTSize}$, $(t + 16) \% \text{HTSize}$, $(t + 17) \% \text{HTSize}$, $(t + 18) \% \text{HTSize}$, $(t + 19) \% \text{HTSize}$, $(t + 20) \% \text{HTSize}$, $(t + 21) \% \text{HTSize}$, $(t + 22) \% \text{HTSize}$, $(t + 23) \% \text{HTSize}$, $(t + 24) \% \text{HTSize}$, $(t + 25) \% \text{HTSize}$, $(t + 26) \% \text{HTSize}$, $(t + 27) \% \text{HTSize}$, $(t + 28) \% \text{HTSize}$, $(t + 29) \% \text{HTSize}$, $(t + 30) \% \text{HTSize}$, $(t + 31) \% \text{HTSize}$, $(t + 32) \% \text{HTSize}$, $(t + 33) \% \text{HTSize}$, $(t + 34) \% \text{HTSize}$, $(t + 35) \% \text{HTSize}$, $(t + 36) \% \text{HTSize}$, $(t + 37) \% \text{HTSize}$, $(t + 38) \% \text{HTSize}$, $(t + 39) \% \text{HTSize}$, $(t + 40) \% \text{HTSize}$, $(t + 41) \% \text{HTSize}$, $(t + 42) \% \text{HTSize}$, $(t + 43) \% \text{HTSize}$, $(t + 44) \% \text{HTSize}$, $(t + 45) \% \text{HTSize}$, $(t + 46) \% \text{HTSize}$, $(t + 47) \% \text{HTSize}$, $(t + 48) \% \text{HTSize}$, $(t + 49) \% \text{HTSize}$, $(t + 50) \% \text{HTSize}$, $(t + 51) \% \text{HTSize}$, $(t + 52) \% \text{HTSize}$, $(t + 53) \% \text{HTSize}$, $(t + 54) \% \text{HTSize}$, $(t + 55) \% \text{HTSize}$, $(t + 56) \% \text{HTSize}$, $(t + 57) \% \text{HTSize}$, $(t + 58) \% \text{HTSize}$, $(t + 59) \% \text{HTSize}$, $(t + 60) \% \text{HTSize}$, $(t + 61) \% \text{HTSize}$, $(t + 62) \% \text{HTSize}$, $(t + 63) \% \text{HTSize}$, $(t + 64) \% \text{HTSize}$, $(t + 65) \% \text{HTSize}$, $(t + 66) \% \text{HTSize}$, $(t + 67) \% \text{HTSize}$, $(t + 68) \% \text{HTSize}$, $(t + 69) \% \text{HTSize}$, $(t + 70) \% \text{HTSize}$, $(t + 71) \% \text{HTSize}$, $(t + 72) \% \text{HTSize}$, $(t + 73) \% \text{HTSize}$, $(t + 74) \% \text{HTSize}$, $(t + 75) \% \text{HTSize}$, $(t + 76) \% \text{HTSize}$, $(t + 77) \% \text{HTSize}$, $(t + 78) \% \text{HTSize}$, $(t + 79) \% \text{HTSize}$, $(t + 80) \% \text{HTSize}$, $(t + 81) \% \text{HTSize}$, $(t + 82) \% \text{HTSize}$, $(t + 83) \% \text{HTSize}$, $(t + 84) \% \text{HTSize}$, $(t + 85) \% \text{HTSize}$, $(t + 86) \% \text{HTSize}$, $(t + 87) \% \text{HTSize}$, $(t + 88) \% \text{HTSize}$, $(t + 89) \% \text{HTSize}$, $(t + 90) \% \text{HTSize}$, $(t + 91) \% \text{HTSize}$, $(t + 92) \% \text{HTSize}$, $(t + 93) \% \text{HTSize}$, $(t + 94) \% \text{HTSize}$, $(t + 95) \% \text{HTSize}$, $(t + 96) \% \text{HTSize}$, $(t + 97) \% \text{HTSize}$, $(t + 98) \% \text{HTSize}$, $(t + 99) \% \text{HTSize}$.

$\text{HTSize} \div 4 (t + 9) \% \text{HTSize}, \dots, (t + i2) \% \text{HTSize}$. That is, the probe sequence is: $t, (t + 1) \% \text{HTSize}, (t + 22) \% \text{HTSize}, (t + 32) \% \text{HTSize}, \dots, (t + i2) \% \text{HTSize}$